

Data Structure (C Programming)

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Roll No : 15

Batch : B1

Prerequisite

Based on Looping Statements.

Q1) Write a C program to print the multiplication table of a number entered by the user using a **for** loop.

Input:

Que 1] Write a C Program to Print the multiplication table of a number entered by the user using a for loop.

```
#include <stdio.h>
int main()
{
    int n, i;
    printf("Enter a Number Total Number by user:");
    scanf("%d", &n);
    for (i = 1; i <= 10; i++)
    {
        printf("%d X %d = %d\n", n, i, (n*i));
    }
    return 0;
}
```

Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ gcc -o output Q1.c
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ ./output
Enter Number:7
The Multiplication of the 7 is
7
14
21
28
35
42
49
56
63
70
```

Q2. Write a C program to calculate the sum of all even numbers entered by user using do..while loop.

Input:

```
#include<stdio.h>
int main()
{
    int n, num, i=1, sum=0;
    printf("Enter Number: ");
    scanf("%d", &n);
    do
    {
        printf("Enter Number: ");
        scanf("%d", &num);
        if(num % 2 == 0)
        {
            sum = sum + num;
        }
        i++;
    } while (i <= n);
    printf("Sum of even numbers = %d", sum);
    return 0;
}
```

O/P:- Enter numbers: 4
Enter number 1: 4
Enter number 2: 3
Enter number 3: 6
Enter number 4: 1
Sum of even numbers = 10.

Output:

```
student@student-OptiPlex-SFF-7020:~$ vi sum.c
student@student-OptiPlex-SFF-7020:~$ gcc -o num sum.c
student@student-OptiPlex-SFF-7020:~$ ./num
Enter numbers4
enter number 1:4
enter number 2:3
enter number 3:6
enter number 4:1
sum of even numbers=10student@student-OptiPlex-SFF-7020:~$
```

Q3. Write a C program to check whether a number is a palindrome or not using a **while** loop.

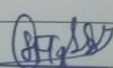
Input:

Q3] Write a C program to check whether a number is a Palindrome or not using a while loop.

```
#include <stdio.h>
int main()
{
    int n, original, rem, reverse = 0;
    printf("Enter a Number:");
    scanf("%d", &n);
    original = n;
    while (n != 0)
    {
        rem = n % 10;
        reverse = reverse * 10 + rem;
        n = n / 10;
    }
    if (original == reverse)
    {
        printf("Number is palindrome");
    }
    else
    {
        printf("Number is not palindrome");
    }
    return 0;
}
```

O/P: Enter a Number: 153.
Number is Palindrome

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Roll No: 15

Sign: 

Output:

```
student@student-OptiPlex-SFF-7020:~$ vi palin.c
student@student-OptiPlex-SFF-7020:~$ gcc -o pal palin.c
student@student-OptiPlex-SFF-7020:~$ ./pal
enter number:121
number is palindromestudent@student-OptiPlex-SFF-7020:~$
```

